

ABCDs of Nutritional Evaluation

Part A: Anthropometrics and Vital Signs



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APPLYING FUNCTIONAL MEDICINE IN CLINICAL PRACTICE

Disclosures

Elizabeth Boham, MD, RD, disclosed no relevant financial relationships with any ineligible companies.

Michael Stone, MD, is a consultant for Tailored Script and Calroy Health.

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Learning Objectives

At the conclusion of this segment, successful participants will be able to:

1. Explain the roles of various anthropometric measurements in the clinical evaluation of cardiometabolic risk.
2. Recall the stepwise approach to body composition assessment.
3. Explain the association between elevated blood pressure and nutrient deficiencies and the therapeutic importance of nutrient repletion.

IFM Clinical Skills and Performance Objectives

THIS SEGMENT WILL ENABLE PARTICIPANTS TO DEVELOP & APPLY THESE SKILLS IN THEIR PRACTICE:

1. Utilize anthropometric measurements and evaluate a patient's body composition to determine cardiometabolic risk profile.

Clinical Focus

- Overview of ABCDs
- Elements of anthropometrics in the functional nutrition assessment.



ABCDs of the Functional Nutrition Evaluation



**Anthropometrics
& Vital Signs**



**Biomarkers &
Functional Labs**

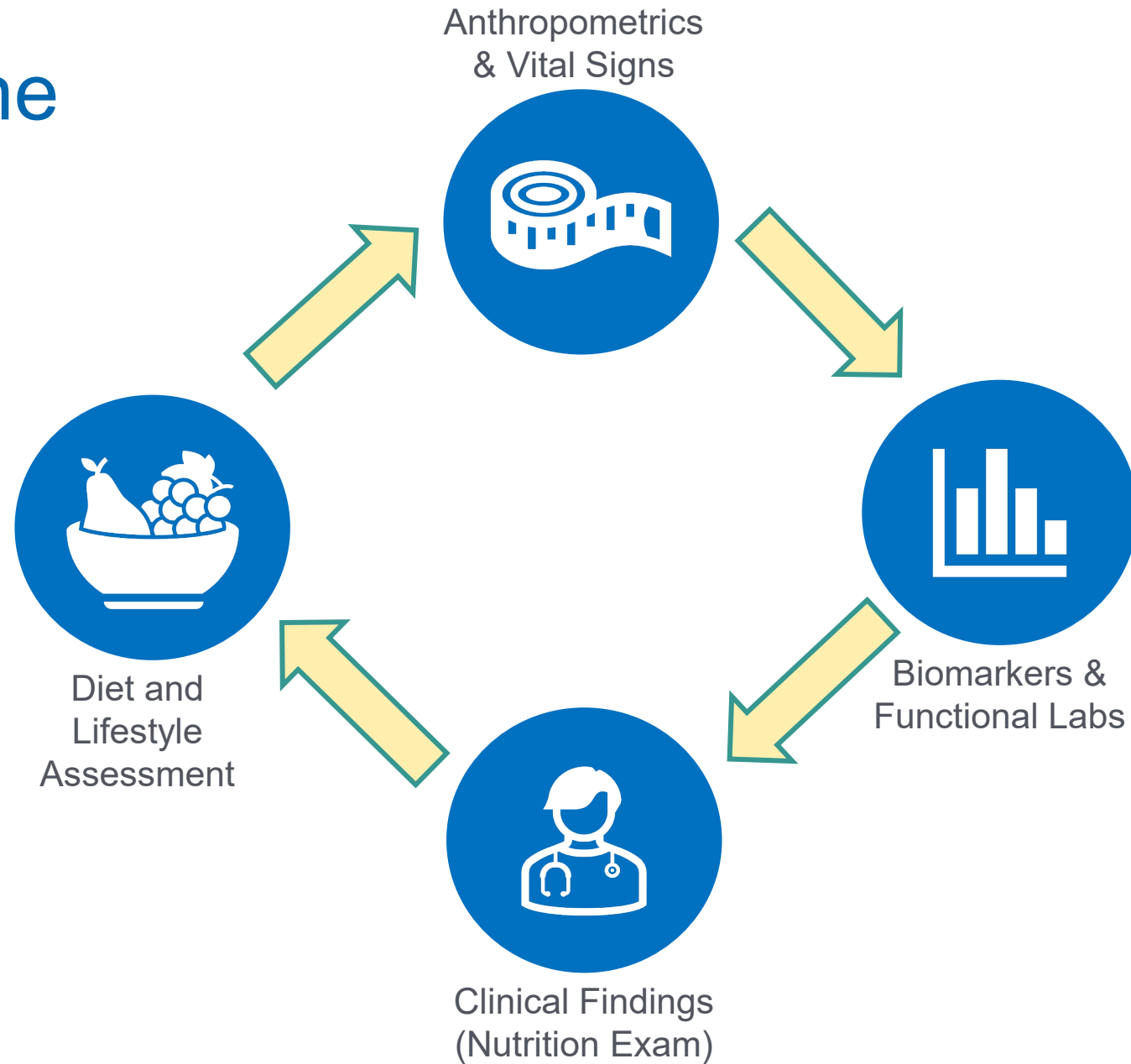


**Clinical Findings
(Nutrition Exam)**

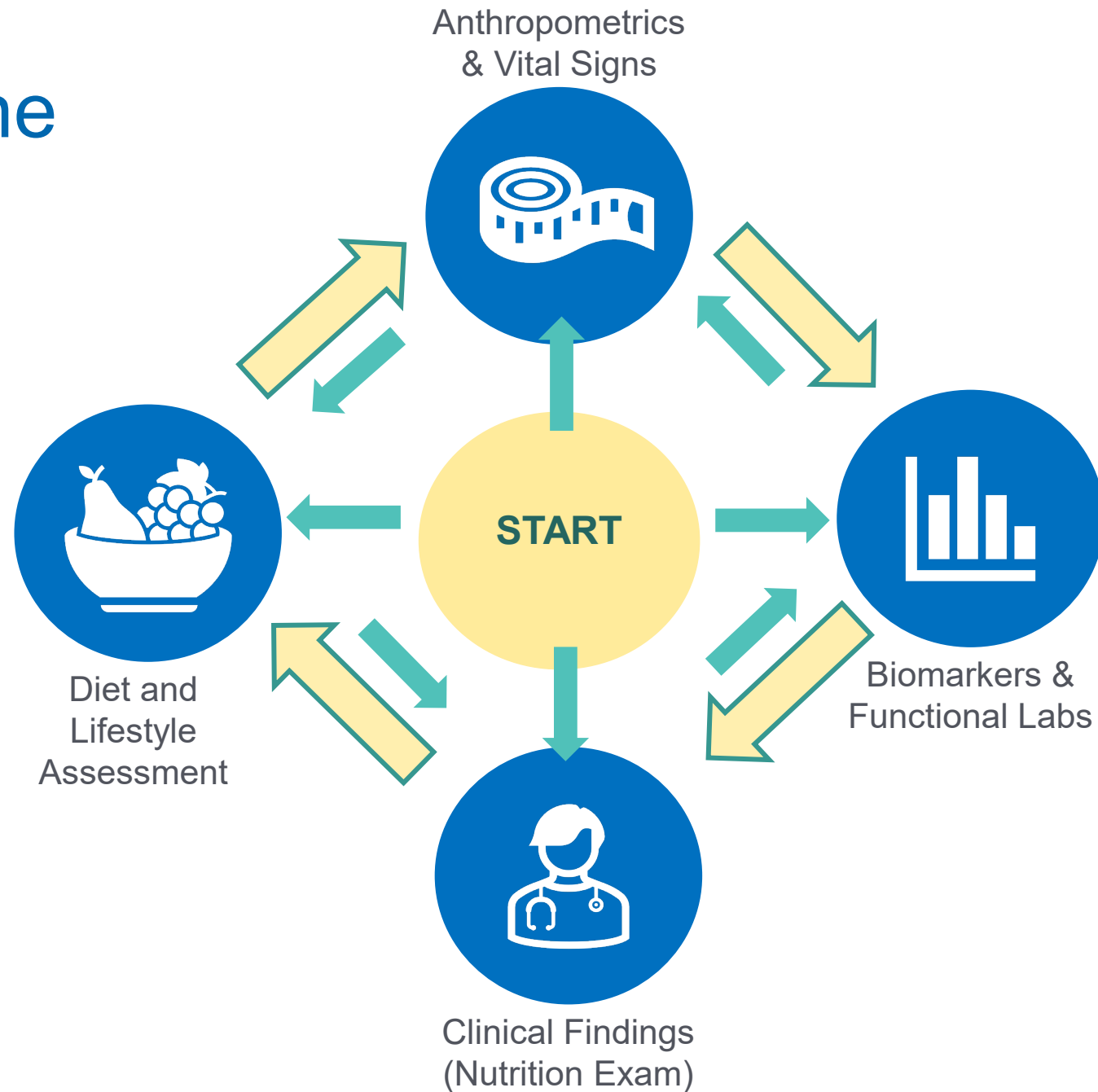


**Diet and Lifestyle
Assessment**

ABCDs of the Functional Nutrition Evaluation



ABCDs of the Functional Nutrition Evaluation



Long Description: A B C Ds of the Functional Nutrition Evaluation

A graphic depicting a sequence:

- **Anthropometrics and Vital Signs:** Involves body measurements and health indicators.
- **Biomarkers and Functional Labs:** Involves testing biological markers for health assessment.
- **Clinical Findings (Nutrition Exam):** Involves observations from a nutrition-focused physical exam.
- **Diet and Lifestyle Assessment:** Involves evaluating dietary habits and lifestyle factors.

The graphic suggests that these elements are interconnected steps in a comprehensive nutritional evaluation process.

Anthropometrics and Vital Signs

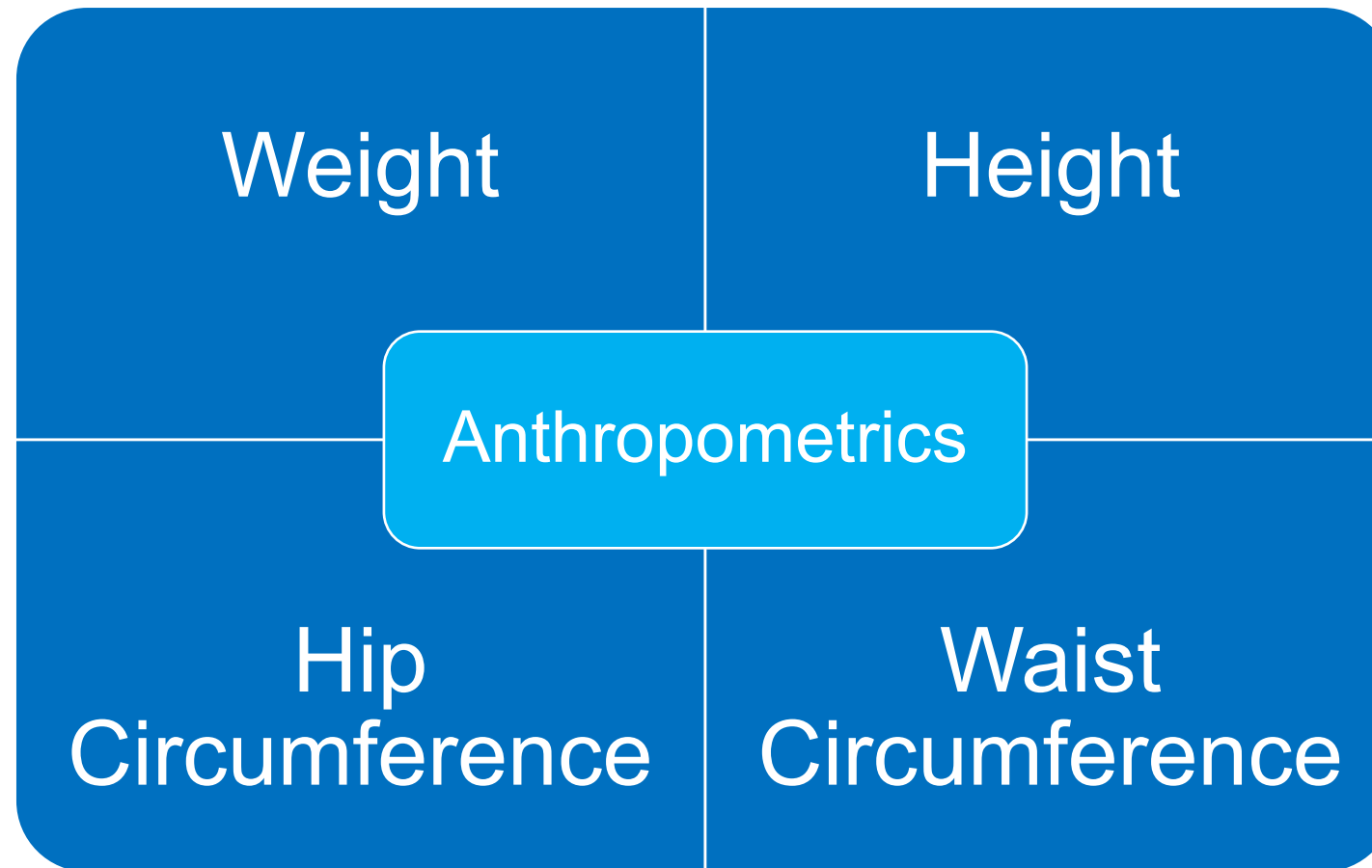
Anthropometrics (Body Composition)

- **Body Mass Index (BMI)**
- **Truncal Obesity Assessment:**
 - Waist circumference (WC)
 - Hip circumference
 - Waist-to-height ratio (WHtR)
 - Waist-to-hip ratio (WHR)
 - Body roundness index (BRI)
- **Body Fat %**
 - Bioelectrical impedance analysis (BIA)
 - Dual energy X-ray absorptiometry (DEXA)

Vital Signs

- **Blood Pressure**

Anthropometric Measurements



Three Ways to Define Metabolic Risk

- Body Mass Index (BMI)
- Weight/Height Ratio (WHtR)

Overweight or Obese



- Waist/Hip Ratio (WHR)
- Body Roundness Index (BRI)
- Waist Circumference (WC)

OverVAT



- Total body fat % may be estimated by other methods
- Bioimpedance Analysis (BIA)
- DEXA

OverFAT



Clinical Focus

- A = Anthropometrics
- BMI and overweight



Anthropometrics and Vital Signs

Anthropometrics (Body Composition)

Vital Signs

- **Body Mass Index (BMI)**
- **Truncal Obesity Assessment:**
 - Waist circumference (WC)
 - Hip circumference
 - Waist-to-height ratio (WHtR)
 - Waist-to-hip ratio (WHR)
 - Body roundness index (BRI)
- **Body Fat %**
 - Bioelectrical impedance analysis (BIA)
 - Dual energy X-ray absorptiometry (DEXA)
- **Blood Pressure**

First Way to Define Risk:

BMI – Is the patient overweight?

- Body Mass Index (BMI)
- Weight/Height Ratio (WHtR)

Overweight or
Obese



- $\text{Weight (lb)} / \text{Height (inches)}^2 \times 703 = \text{BMI}$

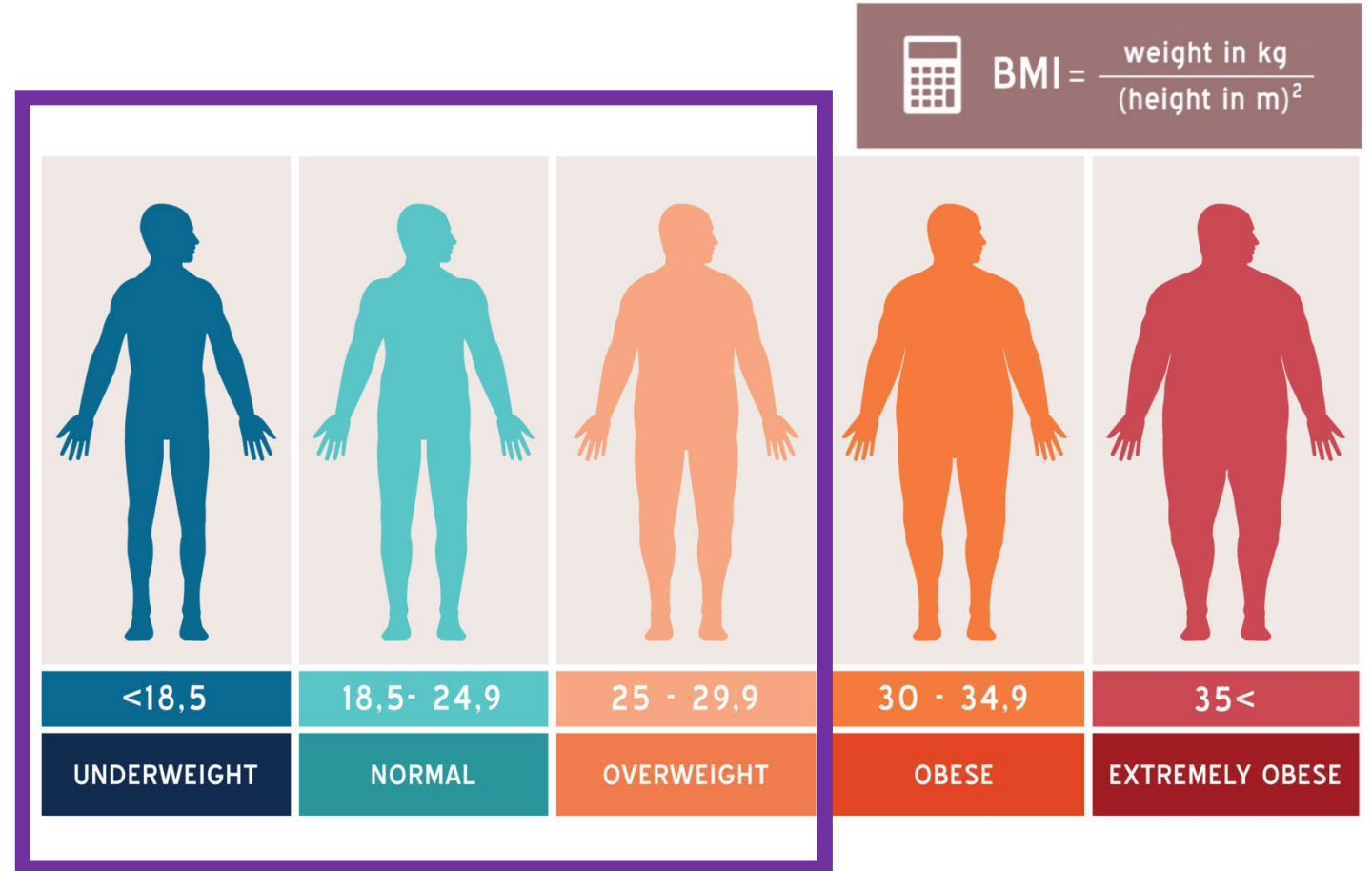
Or

- $\text{Weight (kg)} / \text{Height (m)}^2 = \text{BMI}$
- >25 Overweight, >30 Obese

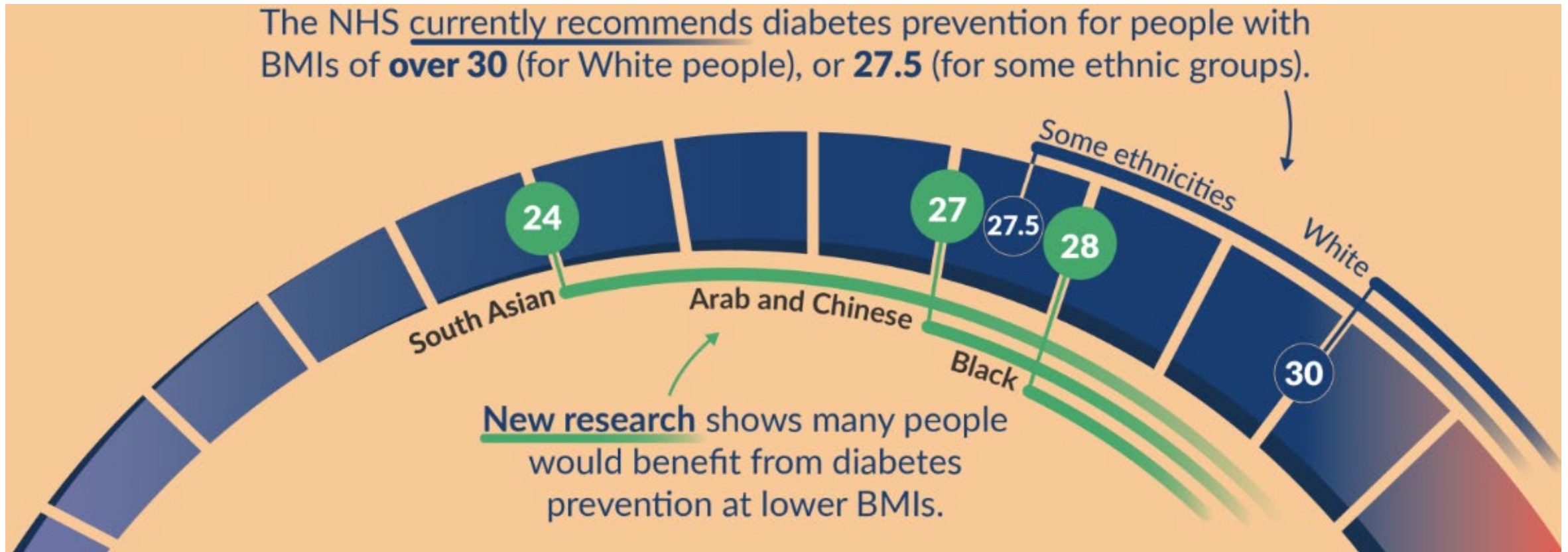
2022: Thresholds for Diabetes Screening by Race and Ethnicity in US Adults >35 Years Old

BMI thresholds:

- **>18.5 kg/m²** → Black and Hispanic Americans
- **>20 kg/m²** → Asian Americans
- **>25 kg/m²** → White Americans



NHS: Thresholds for Diabetes Prevention by Race and Ethnicity (in the UK)



Long Description: N H S: Thresholds for Diabetes Prevention by Race and Ethnicity (in the U K)

A graphic describes the National Health Service (N H S) thresholds for diabetes prevention based on race and ethnicity in the U K. It shows that the N H S currently recommends diabetes prevention for people with body mass index (B M I) values over 30 for White people, or over 27.5 for some other ethnic groups. In addition to White people, it indicates that South Asian, Arab and Chinese, and Black populations tend to benefit from lower B M I thresholds for diabetes prevention. The graphic highlights recent research suggesting many people from diverse ethnic backgrounds could benefit from earlier intervention at lower B M I levels.

Calculator for BMI – Body Mass Index

https://www.nhlbi.nih.gov/health/educational/lose_wt/BMI/bmicalc.htm

The image shows a screenshot of a BMI calculator interface. At the top, there are two tabs: 'STANDARD' and 'METRIC'. Under the 'STANDARD' tab, there are input fields for 'Your Height:' (feet) and 'Your Weight:' (pounds). Below these fields is a button labeled 'Compute BMI'. A large, thick, pink circular arrow graphic is positioned in the center of the interface, pointing clockwise. To the right of the arrow, there is a vertical scale with tick marks. Below the arrow, there is a label 'Your BMI:' followed by an empty input box.

BMI Categories:

- Underweight = <18.5
- Normal weight = 18.5-24.9
- Overweight = 25-29.9
- Obesity = BMI of 30 or greater

The Problem With BMI

**Cannot
differentiate
between
muscle mass
and fat.**

Body Mass Index



Normal Weight Individuals Can Have High Visceral Adipose Tissue (VAT)

- Subjects with a relatively low BMI, such as “metabolically obese” normal-weight individuals, can have gross increases in abdominal visceral fat.

Think “skinny fat.”

- Conversely, those with a high BMI may have very little intra-abdominal visceral fat.

Waist-to-Height Ratio (WHtR)

- Waist (inches) divided by height (inches)
 - A value of <0.5 (ideal) or >0.5 (obese).
 - Ideal is that a person's waist circumference is less than half of their height.
- In some studies, WHtR outperforms BMI when assessing abdominal fat and predicting cardiometabolic risk and cardiovascular disease.
 - Better at predicting “early health risk” than BMI.
 - Does not factor in age, gender, ethnicity.

Clinical Focus

- A = Anthropometrics
- Visceral Adiposity – “OverVAT”
- Waist to Hip Ratio
- BRI: Body Roundness Index (Waist to Height Calculation)



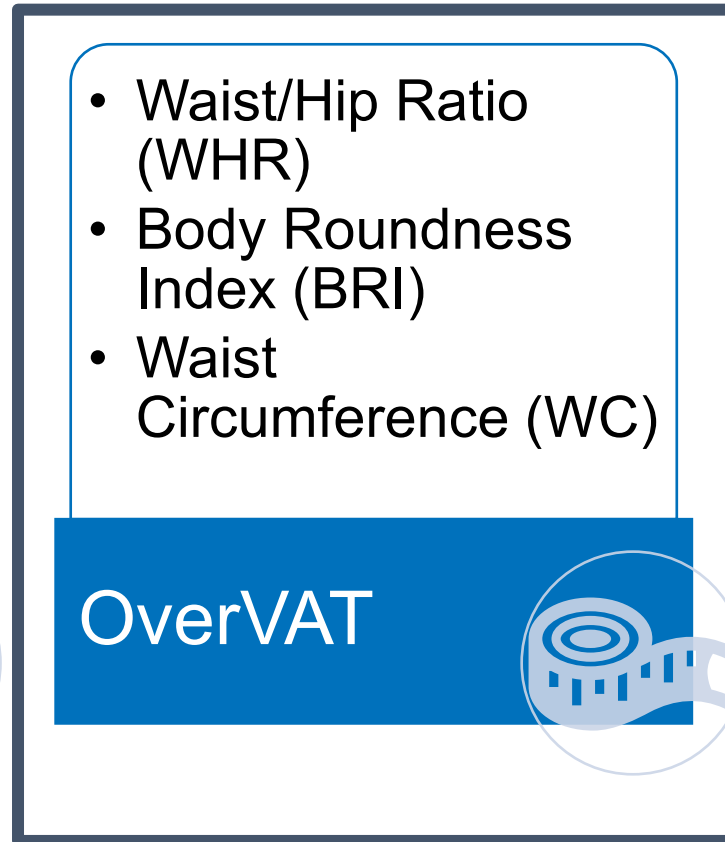
Anthropometrics and Vital Signs

Anthropometrics (Body Composition)

Vital Signs

- **Body Mass Index (BMI)**
- **Truncal Obesity Assessment:**
 - Waist circumference (WC)
 - Hip circumference
 - Waist-to-height ratio (WHtR)
 - Waist-to-hip ratio (WHR)
 - Body roundness index (BRI)
- **Body Fat %**
 - Bioelectrical impedance analysis (BIA)
 - Dual energy X-ray absorptiometry (DEXA)
- **Blood Pressure**

Second Way to Define Risk: Is the patient “OverVAT”?



- Waist circumference is ethnicity- and gender-specific
- Thresholds for greater metabolic risk
- Waist/hip ratio:
>0.85 Female
>0.90 Male

Are We Missing a Good Clue?

Waist circumference is not routinely checked in most offices.



A VITAL SIGN

for insulin and
glycemic
dysregulation

Waist Circumference and Hip Circumference

- Main purpose: to risk-stratify by body shape.
- Helps direct the clinician to possible underlying processes.
- Helps direct biomarker choice.
- Will point toward expectations on clinical exam and diet history and lifestyle patterns.

Concerning Waist Circumference (WC) by Ethnicity

ATP III and International Diabetes Federation

Ethnicity	Circumference – Male	Circumference – Female
USA (ATP III)	40 inches > 102 cm	35 inches > 88 cm
European SS African Mediterranean Middle Eastern	37 inches > 92 cm	31 inches > 80 cm
Asian Hispanic (IDF)	35 inches > 90 cm	31 inches > 80 cm

- Alberti KG, Zimmet P, Shaw J. Metabolic syndrome--a new world-wide definition. A consensus statement from the International Diabetes Federation. *Diabet Med*. 2006;23(5):469-480. doi:10.1111/j.1464-5491.2006.01858.x
- National Heart, Lung, and Blood Institute. *ATP III Guidelines at-a-Glance Quick Desk Reference*. National Institutes of Health; May 2001. Accessed June 8, 2025. <https://www.nhlbi.nih.gov/files/docs/guidelines/at glance.pdf>

Using Waist Circumference (WC)

Table 6 Country/ethnic-specific values for waist circumference

Country/ethnic group	Waist circumference† (as measure of central obesity)	
Europids*	Male	≥ 94 cm
	Female	≥ 80 cm
South Asians‡	Male	≥ 90 cm
	Female	≥ 80 cm
Chinese	Male	≥ 90 cm
	Female	≥ 80 cm
Japanese§	Male	≥ 85 cm
	Female	≥ 90 cm
Ethnic South and Central Americans	Use South Asian recommendations until more specific data are available	
Sub-Saharan Africans	Use European data until more specific data are available	
Eastern Mediterranean and Middle East	Use European data until more specific data are available (Arab) populations	

USA Data



- **Male:** >102 cm/40 inches
- **Female:** >88 cm/35 inches



Waist and Hip Circumference

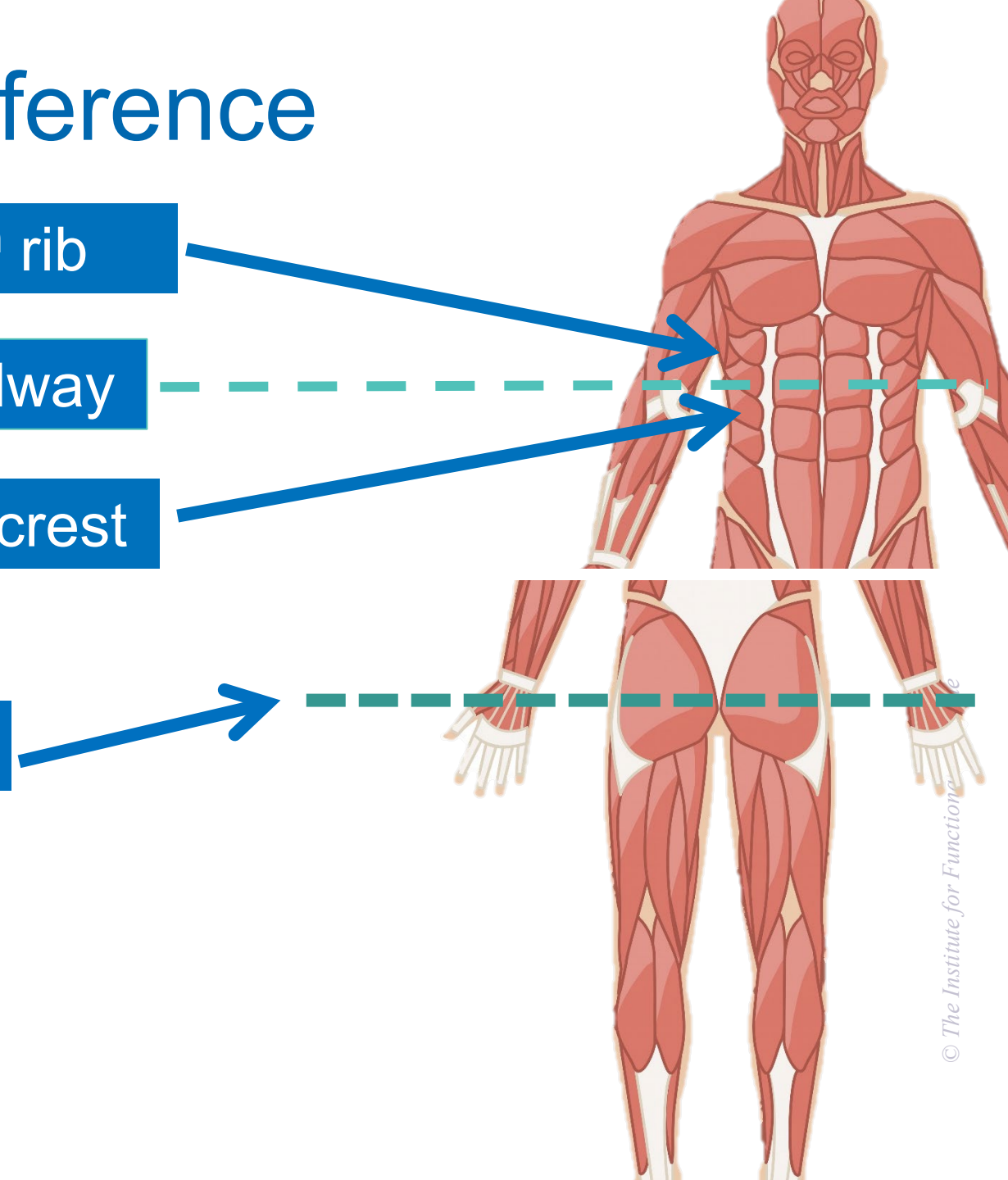
Bottom of the 10th rib

Midway

Top of the iliac crest

Hips over the greater trochanter

Waist (cm)/Hip (cm) =
Waist to Hip Ratio (WHR)



Waist to Hip Ratio (WHR) and Mortality

- A study from the UK Biobank (2006-2022) indicates that WHR has a stronger association with mortality compared to BMI and fat mass index (FMI).
- The association remained across various BMI levels.
- Cutoffs indicating greater metabolic risk (World Health Organization):
 - >0.85 in women
 - >0.90 in men



Body Roundness Index (BRI)

- BRI is a simple, non-invasive option for measuring all-cause mortality risk that accounts for fat distribution and visceral adiposity.
- $BRI = 364.2 - 365.51 - (2\pi WC)^2 / (0.5H)^2$, where WC is waist circumference and H is height, both in meters.
- In the United States, BRI trends increased from 1999-2018, especially among women, people >65 years old, and Mexican Americans.
- BRI has also been identified as a better predictor of hypertension, NAFLD, and cardiovascular disease compared to BMI but comparable to WC.

Calculators for BRI – Body Roundness Index

$$\text{BRI} = 364.2 - 365.51 - (2\pi WC)^2 / (0.5H)^2$$

where WC is waist circumference and H is height, both in meters

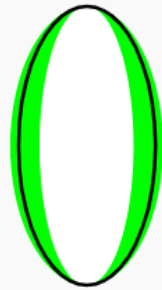
Online calculator options:

- <https://bri-calculator.com/#calculator>
- <https://webfce.com/bri-calculator/> (shown)
 - Option to include hip circumference

BODY ROUNDNESS CALCULATOR

Units ☒ US ☐ Metric	Gender ☒ Female ☐ Male	Ethnicity (Optional) ▼	Age 30	Height 5' 8" - (173 cm) ▼	Weight (lbs) 140
Waist (in) 30		Use hip measure? ☒ Yes ☐ No		Hip (in) 34	

Percent Body Fat: 26.2%
Visceral Adipose Tissue (VAT): 1.2% or 13.4 in³ (Excellent)
Total VAT Mass: 0.4 lbs
Body Roundness Index: 2.3 (in the healthy zone)
Body Mass Index (BMI): 21.2



2.3

■ = Healthy Zone
■ = Individual

[Understanding BRI](#)
[Download PDF](#)

References

BODY ROUNDNESS INDEX (BRI)

- He X, Zhu J, Liang W, et al. Association of body roundness index with cardiovascular disease in patients with cardiometabolic syndrome: a cross-sectional study based on NHANES 2009-2018. *Front Endocrinol (Lausanne)*. 2025;16:1524352. doi:10.3389/fendo.2025.1524352
- Khanmohammadi S, Fallahtafti P, Habibzadeh A, et al. Effectiveness of body roundness index for the prediction of nonalcoholic fatty liver disease: a systematic review and meta-analysis. *Lipids Health Dis*. 2025;24(1):117. doi:10.1186/s12944-025-02544-3
- Wang P, Fan Y, Gao H, Wang B. Body roundness index as a predictor of all-cause and cardiovascular mortality in patients with diabetes and prediabetes. *Diabetes Res Clin Pract*. 2025;219:111958. doi:10.1016/j.diabres.2024.111958
- Zhang X, Ma N, Lin Q, et al. Body roundness index and all-cause mortality among US adults [published correction appears in *JAMA Netw Open*. 2024;7(7):e2426540. doi:10.1001/jamanetworkopen.2024.26540]. *JAMA Netw Open*. 2024;7(6):e2415051. doi:10.1001/jamanetworkopen.2024.15051

Clinical Focus

- A = Anthropometrics
- Total Body Fat %
- “OverFAT”
- BIA – Bioimpedance Analysis
- (Body Roundness Index)



Anthropometrics and Vital Signs

Anthropometrics (Body Composition)

- **Body Mass Index (BMI)**
- **Truncal Obesity Assessment:**
 - Waist circumference (WC)
 - Hip circumference
 - Waist-to-height ratio (WHtR)
 - Waist-to-hip ratio (WHR)
 - Body roundness index (BRI)

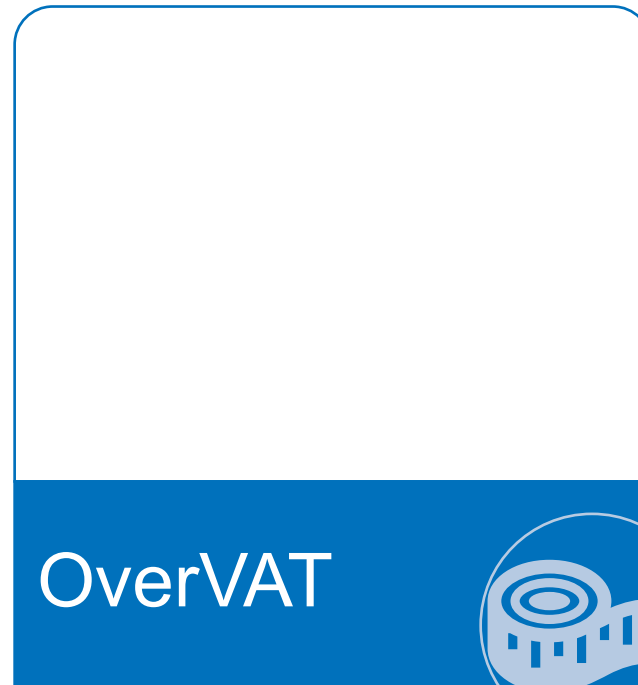
- **Body Fat %**
 - Bioelectrical impedance analysis (BIA)
 - Dual energy X-ray absorptiometry (DEXA)

Vital Signs

- **Blood Pressure**

Third Way to Define Risk

BIA: Is the patient “OverFAT”?



- Total body fat % may be estimated by other methods
- Bioimpedance analysis (BIA)
- DEXA

OverFAT



This complex block contains a list of methods for estimating total body fat percentage, followed by the label 'OverFAT' and an icon of a person. The entire block is enclosed in a dark blue border.

Body Fat %

- Options for Calculating Body Fat %:
 - Circumference Measurements
 - Dual Energy X-Ray Absorptiometry (DEXA)
 - Hydrostatic Weighing
 - Air Displacement Plethysmography
 - Bioimpedance Analysis (BIA)
 - Navy Seal Formula

Bioimpedance Analysis (BIA)

- Hydration Status
- Fluid Distribution:
 - Intracellular water
 - Extracellular water
- Fat Mass
- Lean Body Mass
- Basal Metabolic Rate
- Sarcopenia

BIA and Sarcopenia

- In a systematic review (n=7,668), bioelectrical impedance analysis-derived phase angle (PhA) was found to be **consistently lower in participants with sarcopenia**.
 - Greater cellularity and cell membrane integrity is found with higher values of PhA.
 - PhA is also a predictor of clinical adverse outcomes in elderly patients, including frailty, falls, incident disability, impaired quality of life, and poor muscle strength.

Dual Energy X-Ray Absorptiometry (DEXA)

- Assess bone mineral density
- Assist in the diagnosis of osteoporosis
- Evaluate fat redistribution and visceral adipose tissue
- Evaluate long-term cardiovascular and oncology risk
(due to excessive adiposity)

Body Composition Assessment

Assessment Type	Assess Truncal Obesity	Body Fat %	Assess VAT	Assess Mortality Risk	Notes
Waist Circumference	Yes	Estimate	Yes	Yes	Need to account for ethnicity.
Waist-to-Hip Ratio (WHR)	Yes	Estimate	Yes	Yes	Historical standard; account for gender adiposity distribution; could be used to supplement BMI.
Waist-to-Height Ratio (WHtR)	Yes	Estimate	Yes	Yes	Simple number to track; does not account for age, gender, ethnicity.
BMI	No	Estimate	No	Yes	Does not account for muscle mass, waist circumference.
BRI	Yes	Estimate	Yes	Yes	Emerging standard; includes waist circumference, better mortality risk assessment compared to BMI.
BIA	Yes	Accurate	Yes	Yes	Assesses hydration status, fluid distribution, visceral adiposity, and amount of lean muscle mass.
DEXA	Yes	Accurate	Yes	Yes	Assess bone mineral density; osteoporosis risk; long term cardiovascular & cancer risk.

See: "References: Body Composition Assessment"

References

BODY COMPOSITION ASSESSMENT

- Aggarwal R, Bibbins-Domingo K, Yeh RW, et al. Diabetes screening by race and ethnicity in the United States: equivalent body mass index and age thresholds. *Ann Intern Med*. 2022;175(6):765-773. doi:10.7326/M20-8079
- Alberti KGMM, Zimmet P, Shaw J. Metabolic syndrome—a new world-wide definition. A consensus statement from the International Diabetes Federation. *Diabet Med*. 2006;23(5):469-480. doi:10.1111/j.1464-5491.2006.01858.x
- Centers for Disease Control and Prevention. About adult BMI. Updated March 19, 2024. Accessed June 12, 2025 <https://www.cdc.gov/bmi/adult-calculator/bmi-categories.html>
- Di Vincenzo O, Marra M, Di Gregorio A, Pasanisi F, Scalfi L. Bioelectrical impedance analysis (BIA) -derived phase angle in sarcopenia: a systematic review. *Clin Nutr*. 2021;40(5):3052-3061. doi:10.1016/j.clnu.2020.10.048
- Jayedi A, Soltani S, Zargar MS, Khan TA, Shab-Bidar S. Central fatness and risk of all cause mortality: systematic review and dose-response meta-analysis of 72 prospective cohort studies. *BMJ*. 2020;370:m3324. doi:10.1136/bmj.m3324
- Khalil SF, Mohktar MS, Ibrahim F. The theory and fundamentals of bioimpedance analysis in clinical status monitoring and diagnosis of diseases. *Sensors (Basel)*. 2014;14(6):10895-10928. doi:10.3390/s140610895
- Khan I, Chong M, Le A, et al. Surrogate adiposity markers and mortality [published correction appears in *JAMA Netw Open*. 2025;8(2):e250472. doi:10.1001/jamanetworkopen.2025.0472]. *JAMA Netw Open*. 2023;6(9):e2334836. doi:10.1001/jamanetworkopen.2023.34836

References

BODY COMPOSITION ASSESSMENT

- Marra M, Sammarco R, De Lorenzo A, et al. Assessment of body composition in health and disease using bioelectrical impedance analysis (BIA) and dual energy X-ray absorptiometry (DXA): a critical overview. *Contrast Media Mol Imaging*. 2019;2019:3548284. doi:10.1155/2019/3548284
- Sawicki P, Tałałaj M, Życińska K, Zgliczyński WS, Wierzba W. Current applications and selected technical details of dual-energy X-ray absorptiometry. *Med Sci Monit*. 2021;27:e930839. doi:10.12659/MSM.930839
- Tewari A, Kumar G, Maheshwari A, Tewari V, Tewari J. Comparative evaluation of waist-to-height ratio and BMI in predicting adverse cardiovascular outcome in people with diabetes: a systematic review. *Cureus*. 2023;15(5):e38801. doi:10.7759/cureus.38801
- Thanapholsart J, Khan E, Lee GA. A current review of the uses of bioelectrical impedance analysis and bioelectrical impedance vector analysis in acute and chronic heart failure patients: an under-valued resource? *Biol Res Nurs*. 2023;25(2):240-249. doi:10.1177/10998004221132838
- Wang P, Fan Y, Gao H, Wang B. Body roundness index as a predictor of all-cause and cardiovascular mortality in patients with diabetes and prediabetes. *Diabetes Res Clin Pract*. 2025;219:111958. doi:10.1016/j.diabres.2024.111958
- Zhang X, Ma N, Lin Q, et al. Body roundness index and all-cause mortality among US adults [published correction appears in *JAMA Netw Open*. 2024;7(7):e2426540. doi:10.1001/jamanetworkopen.2024.26540]. *JAMA Netw Open*. 2024;7(6):e2415051. doi:10.1001/jamanetworkopen.2024.15051

ABCDs of Nutritional Evaluation

Part A: Anthropometrics and Vital Signs

Part 2

APPLYING FUNCTIONAL MEDICINE IN CLINICAL PRACTICE

Clinical Focus

- Pattern recognition of body composition and risk.
- The “Body Composition Flow Diagram.”

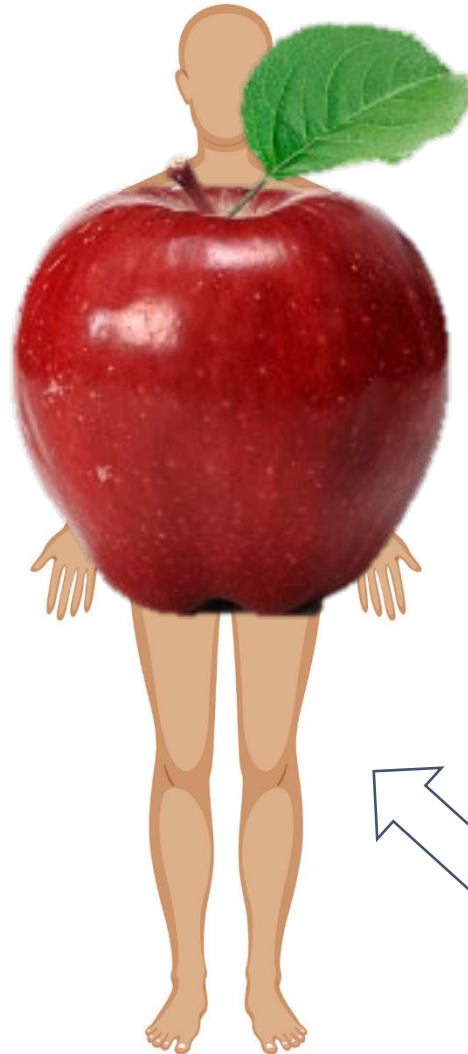


Differentiating Various Altered Body Compositions and Their Associated Risks

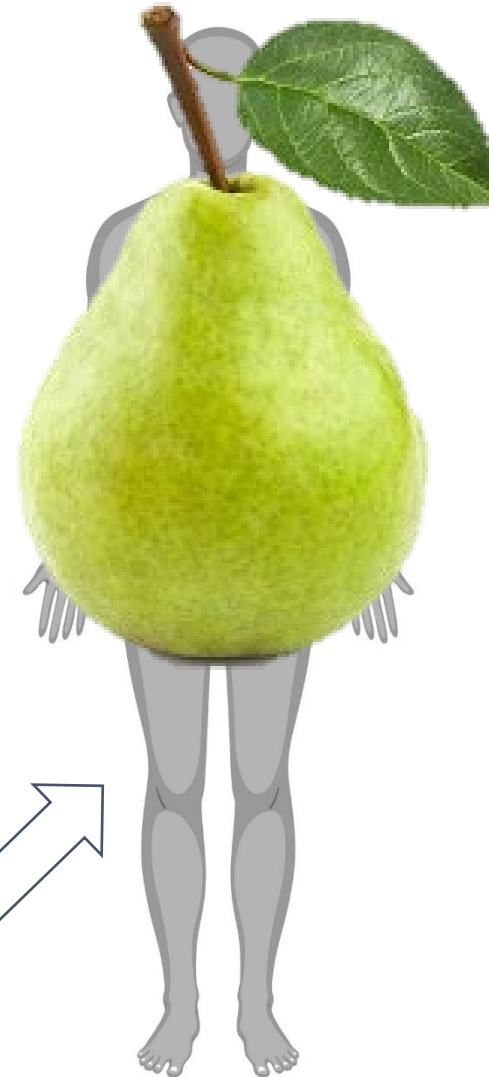
Defining Body Composition

PATTERN RECOGNITION

**Android
Obesity**



**Gynoid
Obesity**



**Overfat
OVER VAT
Visceral
Adipose
Tissue**

**Overfat
OVER SAT
Subcutaneous
Adipose
Tissue**

**OVER
Weight**

In Your IFM Toolkit

Determine:

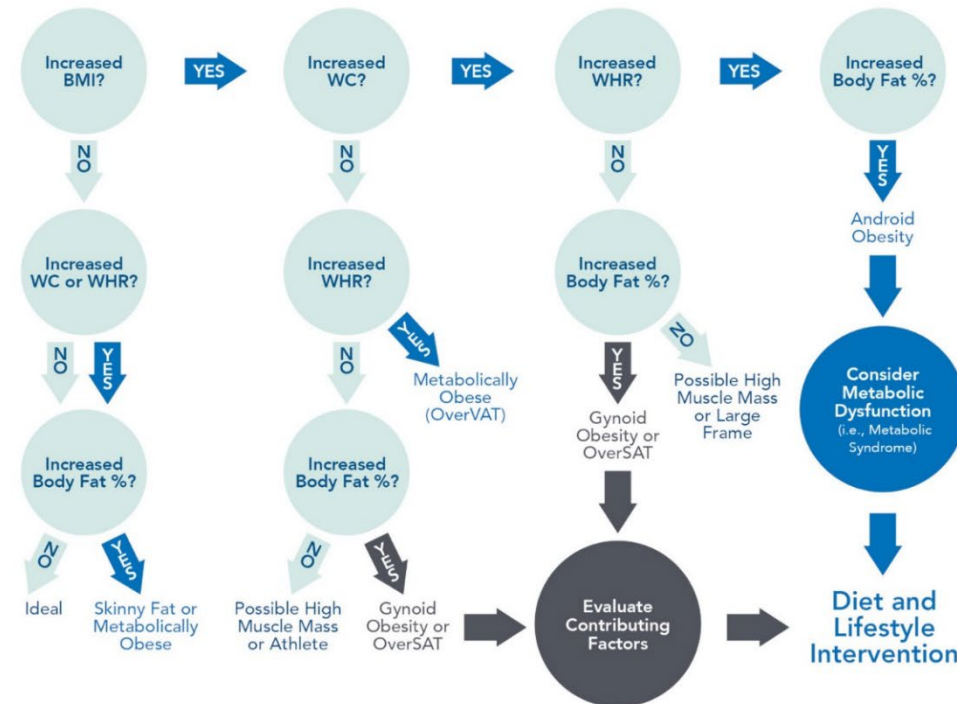
1. BMI
2. Adiposity?
3. Visceral adiposity?



Functional Nutrition Evaluation

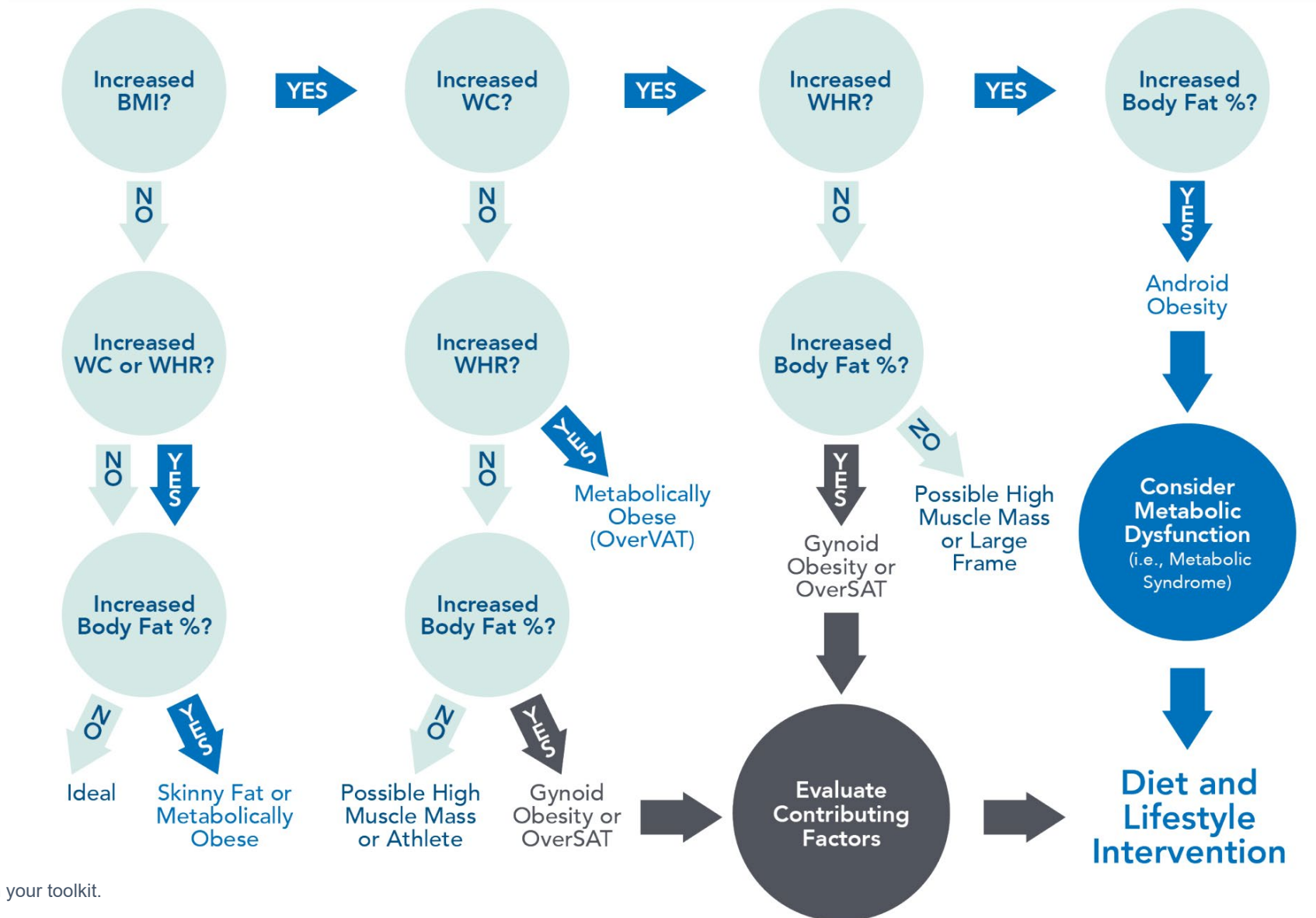
Anthropometrics: Body Composition Flow Diagram

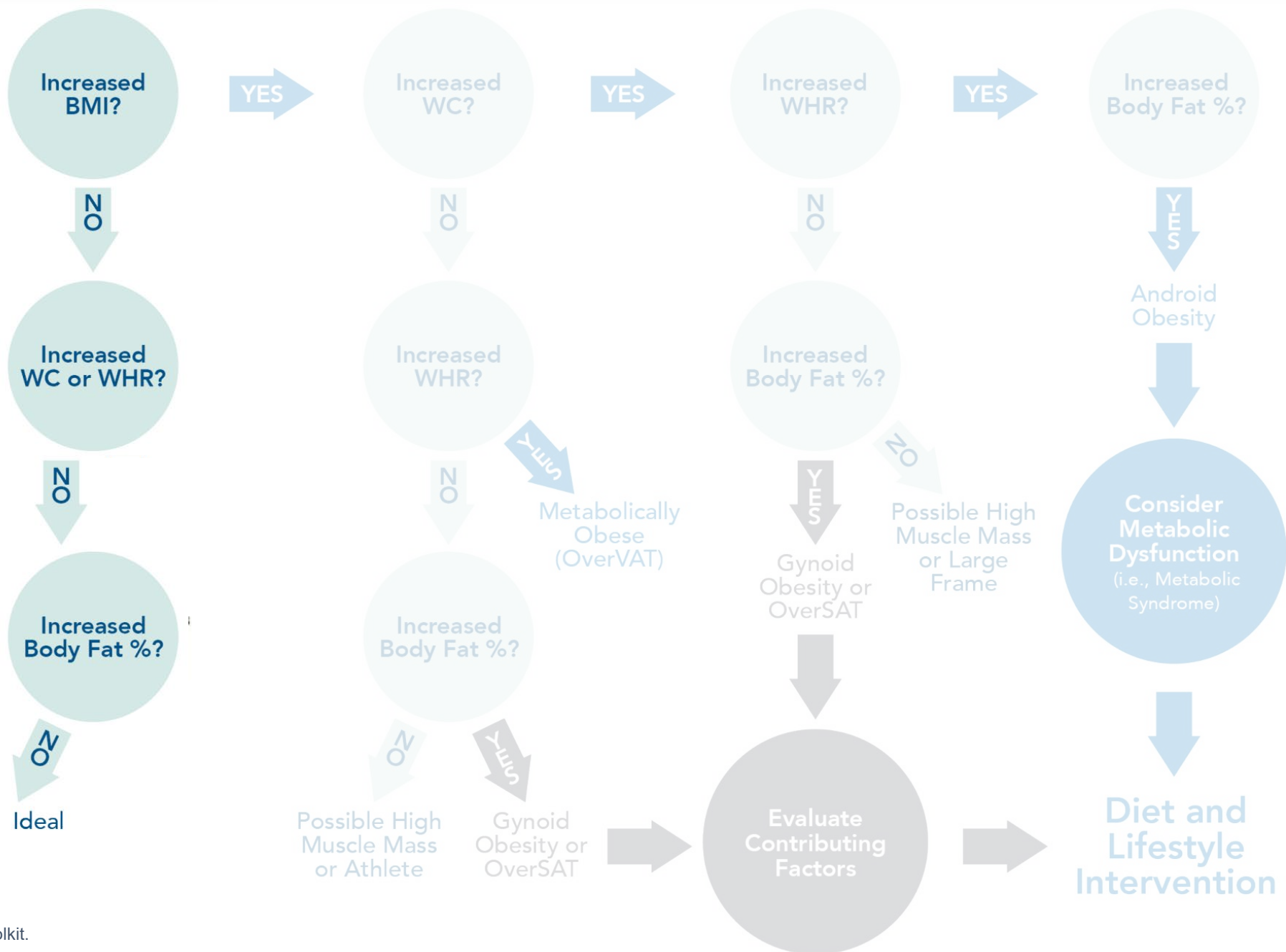
Body composition impacts risk for chronic disease. This diagram can help you assess a patient's disease risk by using anthropometric measurements.

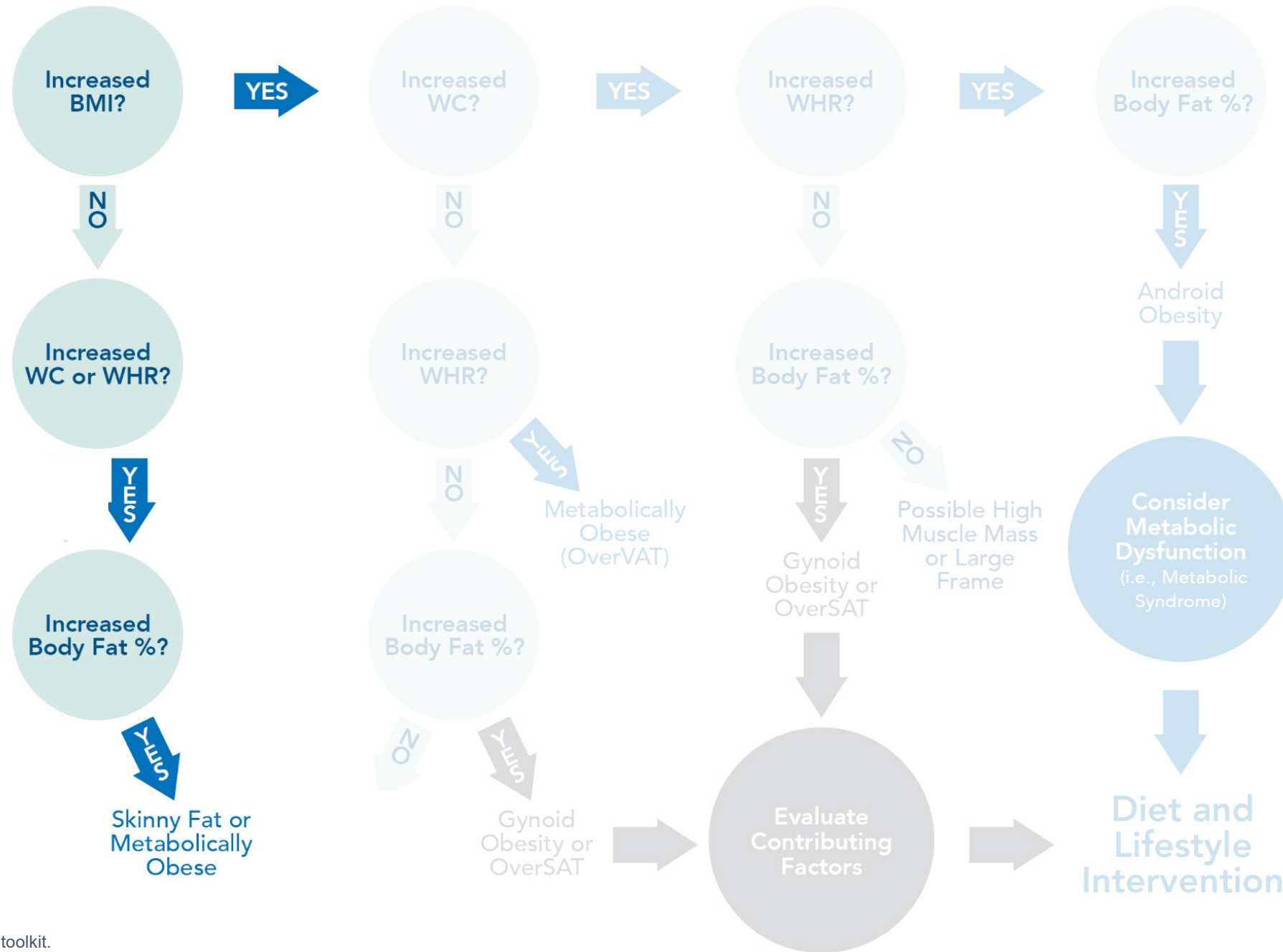


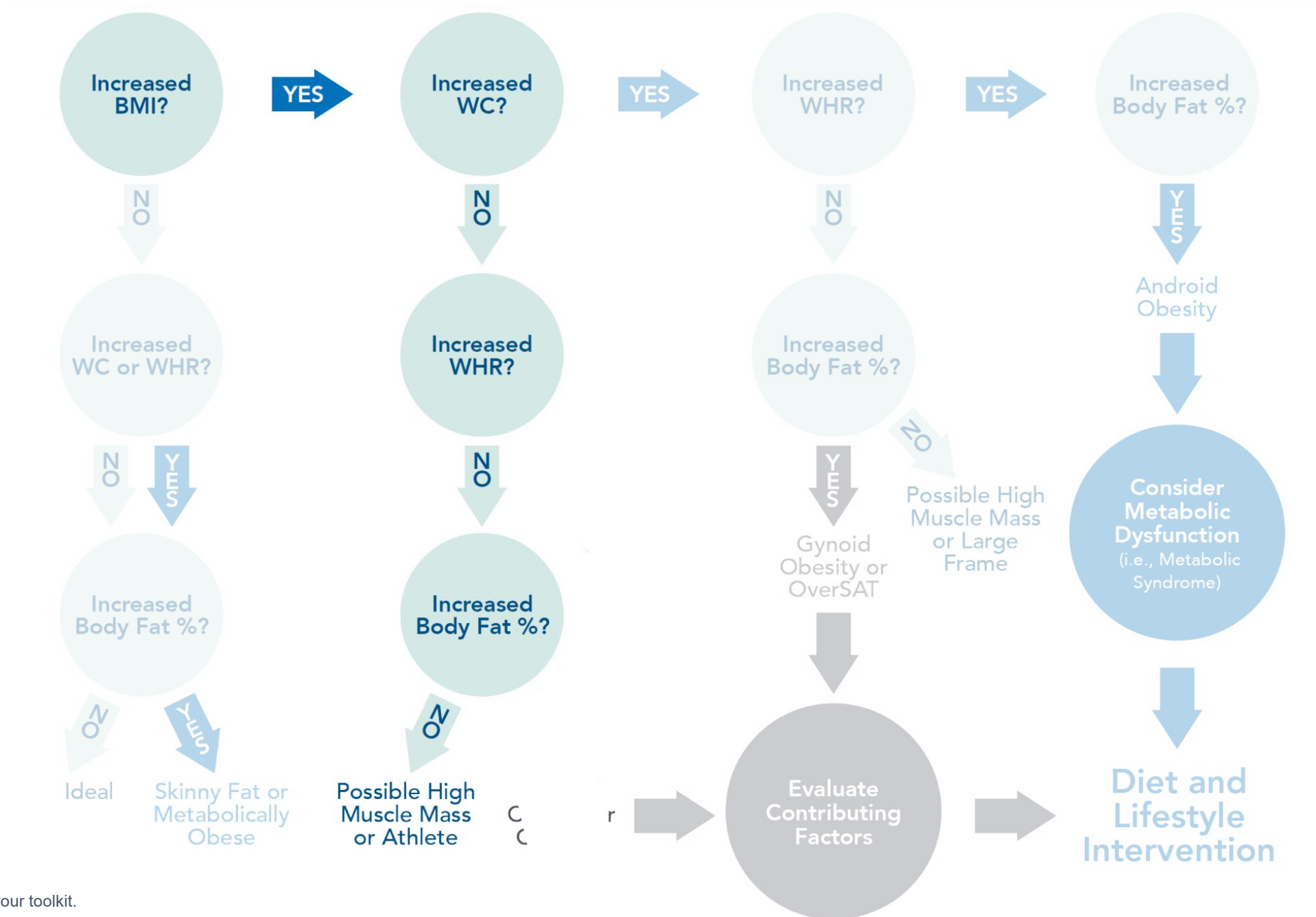
Notes: BMI = body mass index; WC = waist circumference; WHR = waist-to-hip ratio; OverVAT = elevated visceral adipose tissue; OverSAT = elevated subcutaneous adipose tissue

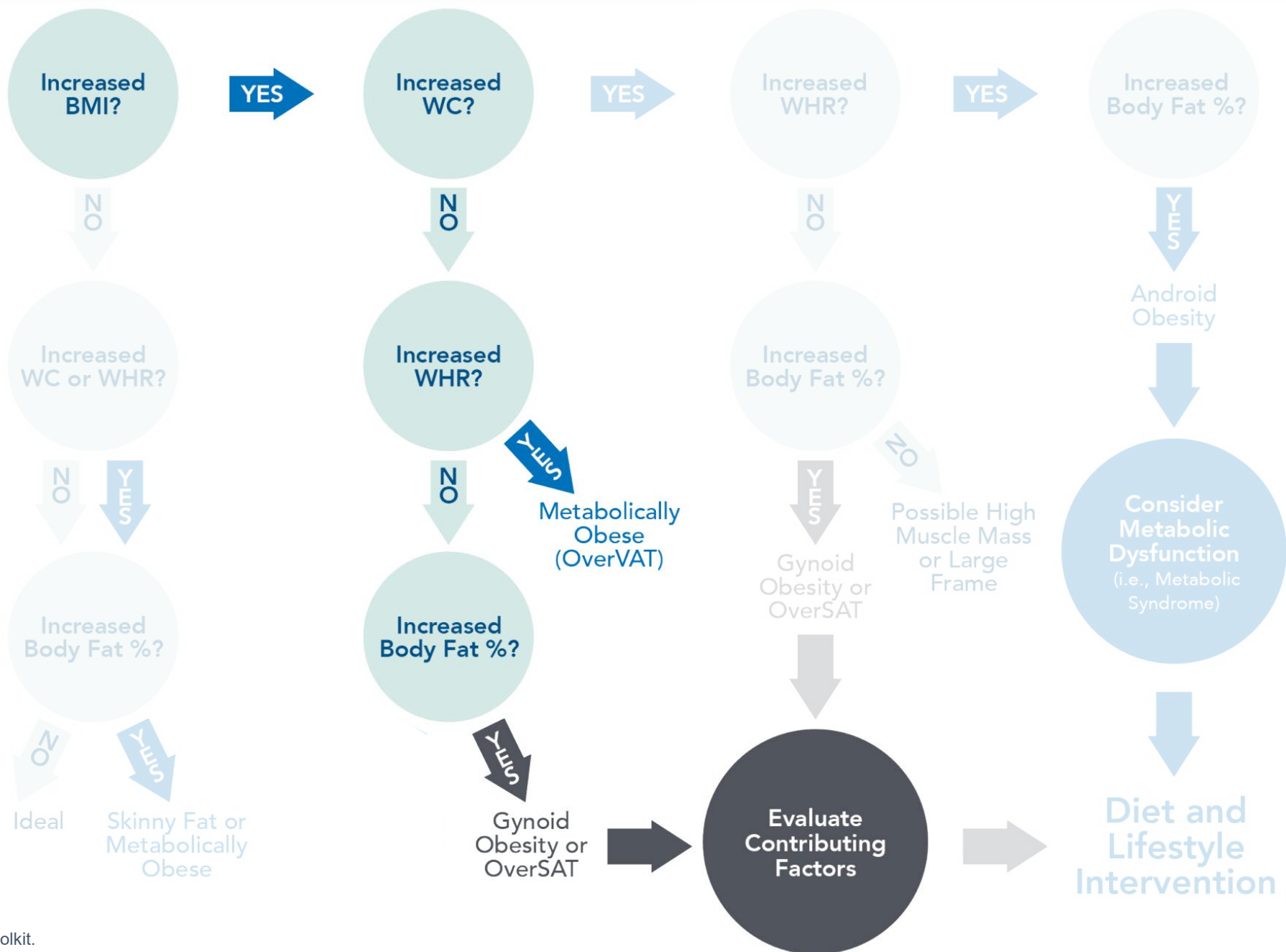
Functional Nutrition Evaluation – Anthropometrics: Body Composition Flow Diagram

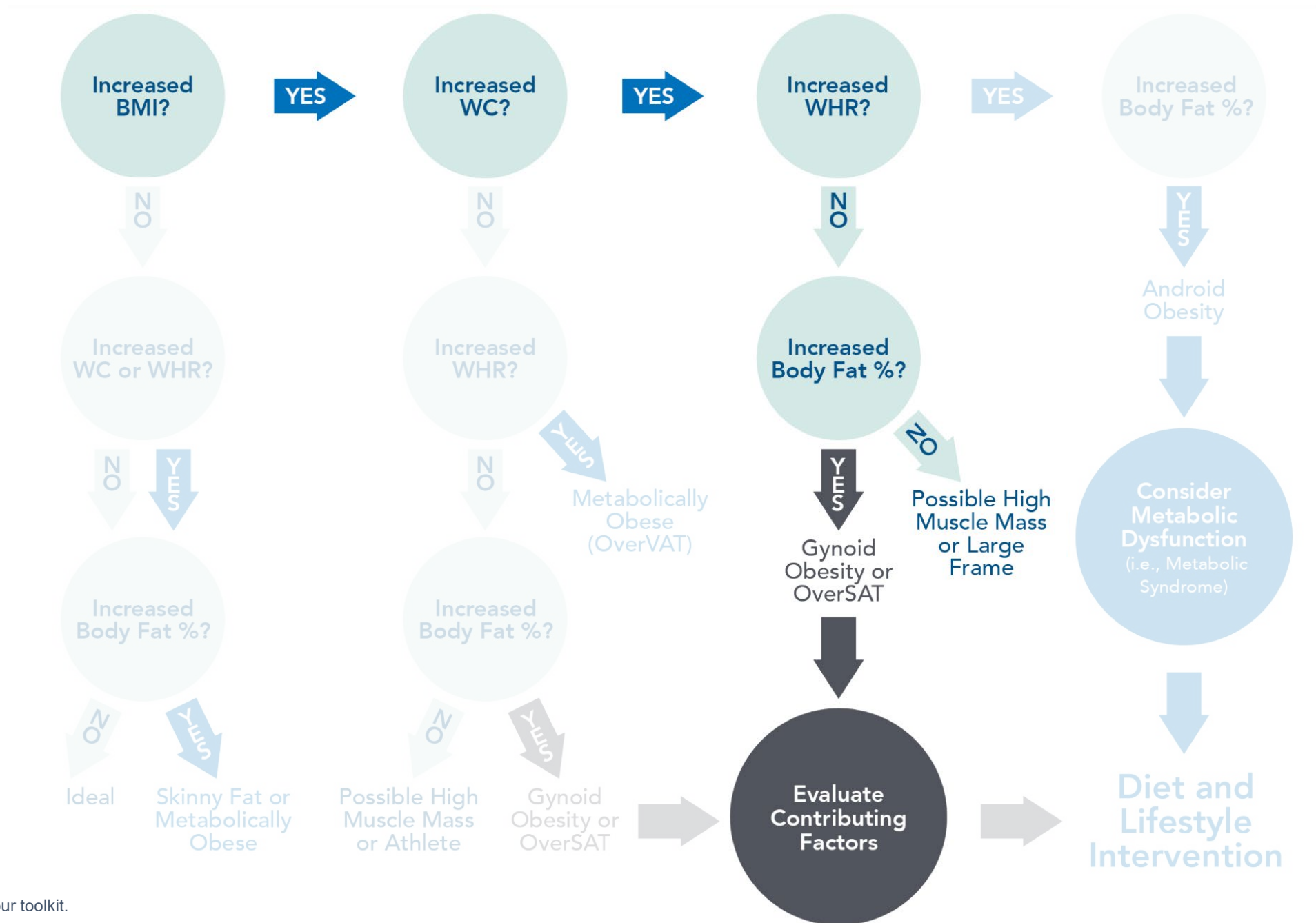


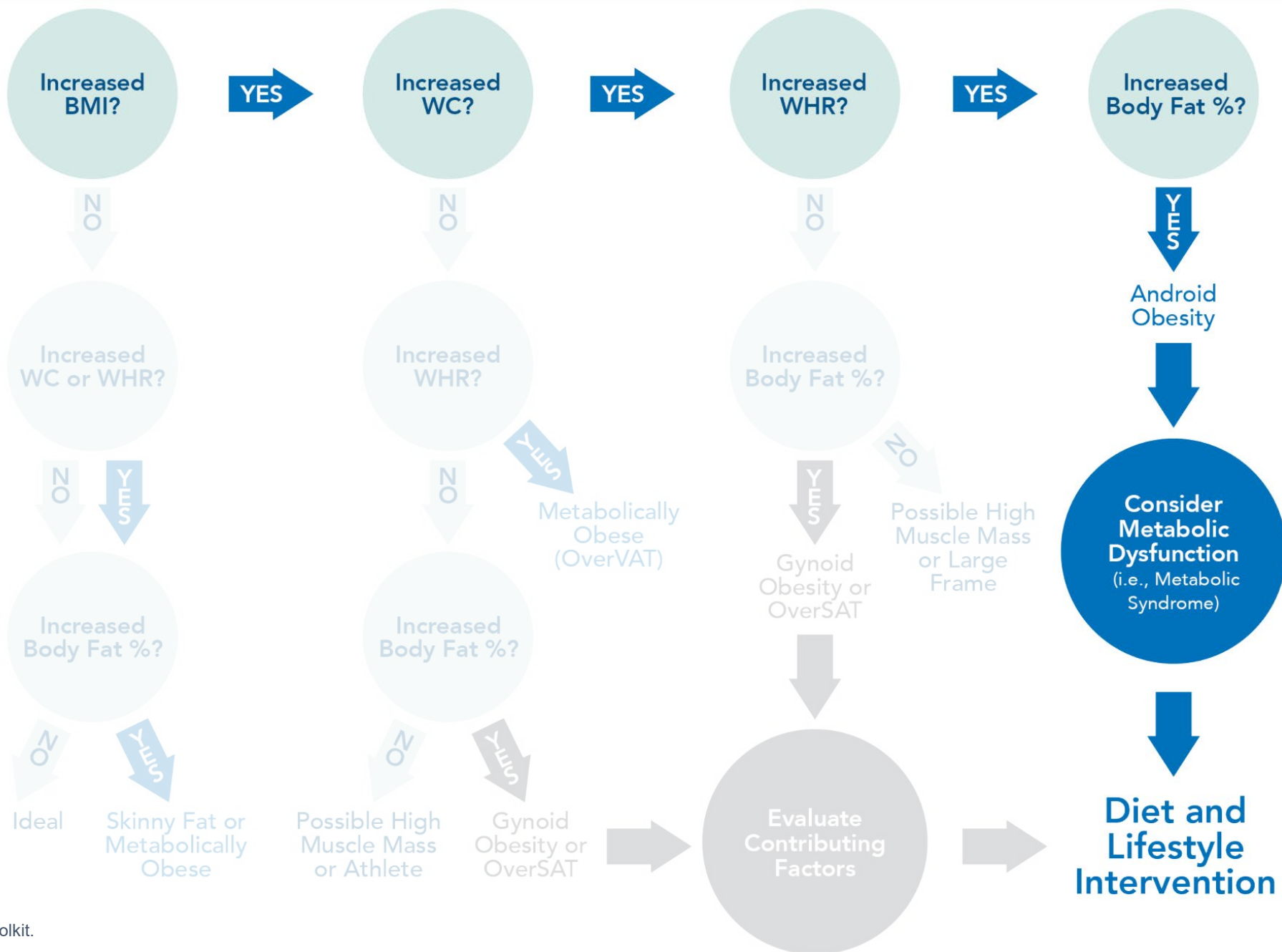


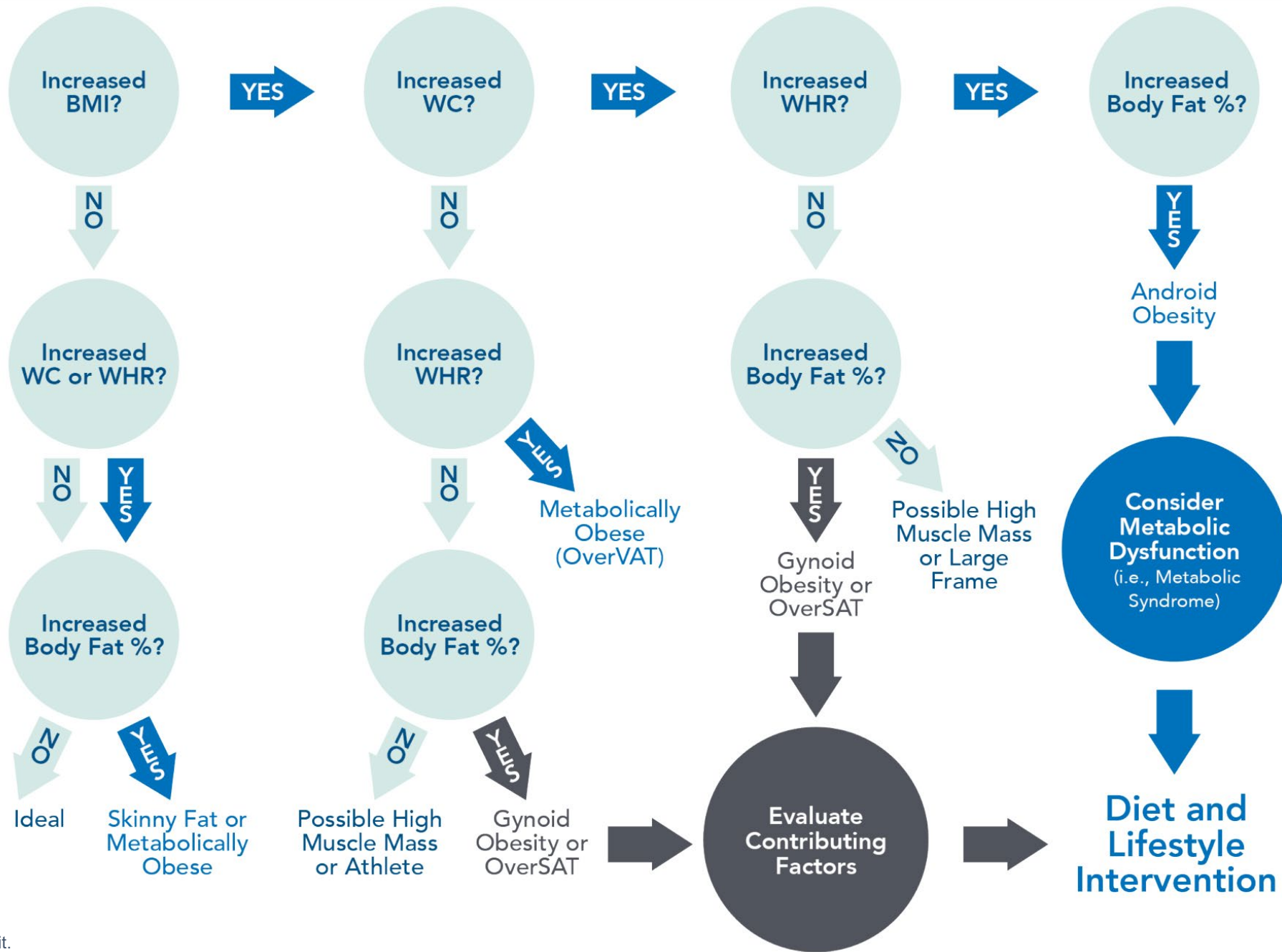












Working the Body Composition

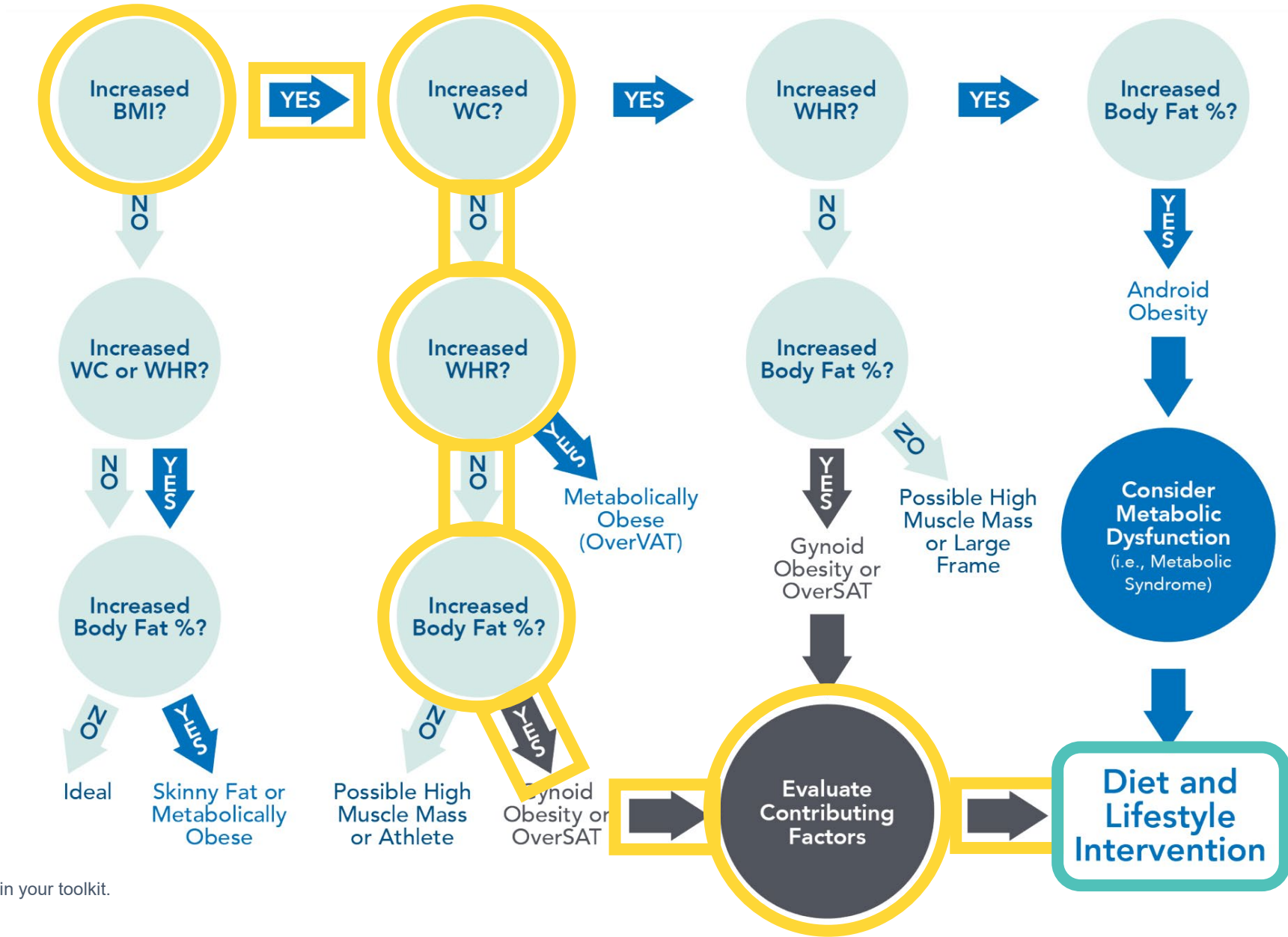
Patient Case

- 34-year-old Hispanic female
- Height: 5'5"
- Weight: 163 lbs
- Waist Circumference: 30 inches (76.2 cm)
- Hip Circumference: 38 inches (96.52 cm)

Working the Body Composition

Assessment Type	Result	Body Fat %	VAT	Mortality Risk	Notes
Waist Circumference	30 inches (76.2 cm)	N/A	Not calculated	Reduced	Patient is within the guidelines based on gender, ethnicity of <31 inches (<80 cm).
Waist-to-Hip Ratio (WHR)	0.79	N/A	Not calculated	Reduced	Within the guidelines of <0.85 in women.
Waist-to-Height Ratio (WHtR)	0.46	N/A	Not calculated	Reduced	Within the guidelines of <0.5.
BMI (NIH calculator)	27.1	N/A	Not calculated	Increased	Patient is classified as “overweight.”
BRI (WebFCE calculator)	2.6	30.5%	2.2% or 32.5 in³ (Average) - Total VAT Mass: 1.1 lbs	Reduced	Within healthy guidelines for BRI and VAT.

Working the Body Composition



References

BODY COMPOSITION ASSESSMENT

- Aggarwal R, Bibbins-Domingo K, Yeh RW, et al. Diabetes screening by race and ethnicity in the United States: equivalent body mass index and age thresholds. *Ann Intern Med*. 2022;175(6):765-773. doi:10.7326/M20-8079
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- Centers for Disease Control and Prevention. About adult BMI. Updated March 19, 2024. Accessed June 12, 2025 <https://www.cdc.gov/bmi/adult-calculator/bmi-categories.html>
- Di Vincenzo O, Marra M, Di Gregorio A, Pasanisi F, Scalfi L. Bioelectrical impedance analysis (BIA) -derived phase angle in sarcopenia: a systematic review. *Clin Nutr*. 2021;40(5):3052-3061. doi:10.1016/j.clnu.2020.10.048
- Jayedi A, Soltani S, Zargar MS, Khan TA, Shab-Bidar S. Central fatness and risk of all cause mortality: systematic review and dose-response meta-analysis of 72 prospective cohort studies. *BMJ*. 2020;370:m3324. doi:10.1136/bmj.m3324
- Khalil SF, Mohktar MS, Ibrahim F. The theory and fundamentals of bioimpedance analysis in clinical status monitoring and diagnosis of diseases. *Sensors (Basel)*. 2014;14(6):10895-10928. doi:10.3390/s140610895
- Khan I, Chong M, Le A, et al. Surrogate adiposity markers and mortality [published correction appears in *JAMA Netw Open*. 2025;8(2):e250472. doi:10.1001/jamanetworkopen.2025.0472]. *JAMA Netw Open*. 2023;6(9):e2334836. doi:10.1001/jamanetworkopen.2023.34836

References

BODY COMPOSITION ASSESSMENT

- Marra M, Sammarco R, De Lorenzo A, et al. Assessment of body composition in health and disease using bioelectrical impedance analysis (BIA) and dual energy X-ray absorptiometry (DXA): a critical overview. *Contrast Media Mol Imaging*. 2019;2019:3548284. doi:10.1155/2019/3548284
- Sawicki P, Tałałaj M, Życińska K, Zgliczyński WS, Wierzba W. Current applications and selected technical details of dual-energy X-ray absorptiometry. *Med Sci Monit*. 2021;27:e930839. doi:10.12659/MSM.930839
- Tewari A, Kumar G, Maheshwari A, Tewari V, Tewari J. Comparative evaluation of waist-to-height ratio and BMI in predicting adverse cardiovascular outcome in people with diabetes: a systematic review. *Cureus*. 2023;15(5):e38801. doi:10.7759/cureus.38801
- Thanapholsart J, Khan E, Lee GA. A current review of the uses of bioelectrical impedance analysis and bioelectrical impedance vector analysis in acute and chronic heart failure patients: an under-valued resource? *Biol Res Nurs*. 2023;25(2):240-249. doi:10.1177/10998004221132838
- Wang P, Fan Y, Gao H, Wang B. Body roundness index as a predictor of all-cause and cardiovascular mortality in patients with diabetes and prediabetes. *Diabetes Res Clin Pract*. 2025;219:111958. doi:10.1016/j.diabres.2024.111958
- Zhang X, Ma N, Lin Q, et al. Body roundness index and all-cause mortality among US adults [published correction appears in *JAMA Netw Open*. 2024;7(7):e2426540. doi:10.1001/jamanetworkopen.2024.26540]. *JAMA Netw Open*. 2024;7(6):e2415051. doi:10.1001/jamanetworkopen.2024.15051

ABCDs of Nutritional Evaluation

Part A: Anthropometrics and Vital Signs

Part 3

APPLYING FUNCTIONAL MEDICINE IN CLINICAL PRACTICE

Clinical Focus

- Blood pressure: A vital sign that is potentially indicative of nutrient deficiencies.



Anthropometrics and Vital Signs

Anthropometrics (Body Composition)

- **Body Mass Index (BMI)**
- **Truncal Obesity Assessment:**
 - Waist circumference (WC)
 - Hip circumference
 - Waist-to-height ratio (WHtR)
 - Waist-to-hip ratio (WHR)
 - Body roundness index (BRI)
- **Body Fat %**
 - Bioelectrical impedance analysis (BIA)
 - Dual energy X-ray absorptiometry (DEXA)

Vital Signs

- **Blood Pressure**

Vital Signs

- BP, HR, RR, temperature, oxygen saturation
- What is taught in school?
- What is historically done with that information?
- Do we look at vital signs differently through a functional medicine lens?
- **Which vital sign is most indicative of nutrient deficiencies?**

2020 Cochrane Review

For the general population of people with elevated blood pressure, **the benefits of achieving a lower blood pressure versus a standard target ($\leq 140/90$ mmHg) do not outweigh the harms associated with this goal.**

Hypertension Guidelines 2017 AHA

Blood pressure categories in the guidelines are:

- **Normal:** Less than 120/80 mm Hg
- **Elevated:** Systolic between 120-129 *and* diastolic less than 80
- **Stage 1:** Systolic between 130-139 *or* diastolic between 80-89
- **Stage 2:** Systolic at least 140 *or* diastolic at least 90 mm Hg
- **Hypertensive crisis:** Systolic over 180 and/or diastolic over 120, with patients needing prompt changes in medication if there are no other indications of problems or immediate hospitalization if there are signs of organ damage.

	BLOOD PRESSURE CATEGORY	SYSTOLIC		DIASTOLIC
	Normal	< 120	and	< 80
	Elevated	120 - 129	and	< 80
	High (Hypertension Stage 1)	130 - 139	or	80 - 89
	High (Hypertension Stage 2)	≥140	or	≥90
	Hypertensive Crisis (immediate intervention)	>180	and/or	> 120

Blood Pressure Targets by Guidelines

Guideline	Year	Blood Pressure Target	
American Academy of Family Physicians	2022	<140/90	
ACC/AHA Guidelines	2017	<130/80	
American College of Physicians and American Academy of Family Physicians	2017	<140/90	<150/90 for patients >60 yo
European Society of Hypertension Council	2024	<140/90	
National Institute for Health and Care Excellence	2023	<135/85	<145/85 for patients >80 yo
International Society of Hypertension	2020	<130/80	<140/90 for patients >60 yo
US Department of Veterans Affairs/US Department of Defense	2020	<130/90	<150/90 for patients >60 yo
Eighth Joint National Committee	2014	<140/90	<150/90 for patients >60 yo

See: "References: Blood Pressure Targets by Guidelines"

References

BLOOD PRESSURE TARGETS BY GUIDELINES

- Armstrong C. High blood pressure: ACC/AHA releases updated guideline. *Am Fam Physician*. 2018;97(6):413-415.
- National Institute for Health and Care Excellence. Hypertension in adults: diagnosis and management: NICE guideline, No. 136. Updated November 21, 2023. Accessed June 8, 2025. <https://www.ncbi.nlm.nih.gov/books/NBK547161/>
- James PA, Oparil S, Carter BL, et al. 2014 evidence-based guideline for the management of high blood pressure in adults: report from the panel members appointed to the Eighth Joint National Committee (JNC 8) [published correction appears in *JAMA*. 2014;311(17):1809]. *JAMA*. 2014;311(5):507-520. doi:10.1001/jama.2013.284427
- McEvoy JW, McCarthy CP, Bruno RM, et al. 2024 ESC guidelines for the management of elevated blood pressure and hypertension [published correction appears in *Eur Heart J*. 2025;46(14):1300. doi:10.1093/eurheartj/ehaf031]. *Eur Heart J*. 2024;45(38):3912-4018. doi:10.1093/eurheartj/ehae178
- News Staff, AAFP News. AAFP issues new clinical practice guideline on hypertension. *Ann Fam Med*. 2023;21(2):190-191. doi:10.1370/afm.2972
- International Society of Hypertension. ISH 2020 global practice guidelines. Published 2020. Accessed June 8, 2025. <https://ish-world.com/global-hypertension-practice-guidelines>
- Department of Veterans Affairs/Department of Defense. VA/DoD Clinical practice guideline for the diagnosis and management of hypertension in the primary care setting. Published 2020. Accessed June 8, 2025. <https://www.healthquality.va.gov/guidelines/CD/htn/VADODHypertensionCPG508Corrected792020.pdf>

Functional Nutrition Evaluation – Anthropometrics: Blood Pressure

DETERMINATION OF BLOOD PRESSURE FROM THE FUNCTIONAL MEDICINE PERSPECTIVE

- Blood pressure determination, part of the ABCDs of functional nutrition evaluation, allows the clinician to consider **the root cause of endothelial dysfunction associated with inflammation, immune dysfunction, and oxidative stress imbalances**. The correct determination of blood pressure and appropriate interventions can change the trajectory of health and disease.
- A comprehensive program that includes **diet and nutrition and optimal exercise to influence healthful weight and body composition** changes can have beneficial impacts on reduction of blood pressure.

• Blumenthal JA, Babyak MA, Hinderliter A, et al. Effects of the DASH diet alone and in combination with exercise and weight loss on blood pressure and cardiovascular biomarkers in men and women with high blood pressure: the ENCORE study. *Arch Intern Med*. 2010;170(2):126-135. doi:10.1001/archinternmed.2009.470

• Kucharska A, Gajewska D, Kiedrowski M, et al. The impact of individualized nutritional therapy according to DASH diet on blood pressure, body mass, and selected biochemical parameters in overweight/obese patients with primary arterial hypertension: a prospective randomised study. *Kardiol Pol*. 2018;76(1):158-165. doi:10.5603/KP.a2017.0184

Association of Nutrient Deficiencies With Hypertension

- **Hypertensive patients had more micronutrient deficiencies.**
- Repeat testing at six months showed significantly improved antioxidant profile by 8.47% ($p=0.03$) and over 97% complete repletion rate of micronutrients.
- **62%** of the hypertensive patients (range 4-12 months) **tapered and discontinued all antihypertensive drugs** with controlled BP of 120/80 mm Hg to 126/84 mm Hg.

Association of Nutrient Deficiencies With Hypertension

Proteins	Fats	Carbohydrates	Minerals	Vitamins	Phytonutrients
• Animal protein (lean or wild) • Plant protein	• Omega-3 fatty acids • Omega-9 fatty acids	• Soluble fiber	• Calcium • Magnesium • Potassium • Zinc	• Vitamin B6 • Vitamin C • Vitamin D • Choline	• Alpha lipoic acid • CoQ10 • Flavonoids • Lycopene • Low antioxidant profile

- Houston MC. The role of nutrition, nutraceuticals, vitamins, antioxidants, and minerals in the prevention and treatment of hypertension. *Altern Ther Health Med*. 2013;19(Suppl 1):32-49.
- Houston M. Nutrition and nutraceutical supplements for the treatment of hypertension: part II. *J Clin Hypertens (Greenwich)*. 2013;15(11):845-851. doi:10.1111/jch.12212
- Houston MC. The role of cellular micronutrient analysis, nutraceuticals, vitamins, antioxidants and minerals in the prevention and treatment of hypertension and cardiovascular disease. *Ther Adv Cardiovasc Dis*. 2010;4(3):165-183. doi:10.1177/1753944710368205

Nutrients and Hypertension

If blood pressure is not improving...
repleting nutrient deficiencies
may be the key to addressing the
underlying root causes.

Anthropometrics: Blood Pressure

Check out the Quick
Reference Guide in
your toolkit.



Functional Nutrition Evaluation

Anthropometrics: Blood Pressure Quick Reference Guide

A person's age, smoking history, pulse pressure, dyslipidemia, and other clinical variables, along with any signs of subclinical organ damage can further increase the associated risk of elevated blood pressure and influence the degree and aggressiveness of needed interventions. The grades of concern are determined based on the blood pressure reading and number of risk factors. Normal and abnormal elevations in blood pressure are highlighted in the table below.

Blood Pressure (mmHg) Risk Assessment

Risk Factors or Disease	Normal: SBP under 120 and DBP under 80	Elevated: SBP 120-129 and DBP under 80	Stage 1 HTN: SBP 130-139 or DBP 80-89	Stage 2 HTN: SBP 140 and higher or DBP 90 and higher	Hypertensive Crisis: SBP over 180 or DBP over 120
No Other Risk Factors	Average Risk — No BP intervention needed	Low Added Risk — Lifestyle changes, exercise and diet interventions	Moderate Added Risk — Lifestyle changes for several months then drug treatment if BP uncontrolled	High Added Risk — Lifestyle changes and consider drug treatment	Prompt adjustments in medication needed
1-2 Risk Factors	Low Added Risk — Lifestyle changes, exercise and diet interventions	Moderate Added Risk — Lifestyle changes for several months then drug treatment if BP uncontrolled	High Added Risk — Lifestyle changes and consider drug treatment	Very High Added Risk — Lifestyle changes and immediate drug treatment	Consider immediate hospitalization
Over 3 Risk Factors, Metabolic Syndrome or Other Disease	Moderate Added Risk — Lifestyle changes for several months then drug treatment if BP uncontrolled	High Added Risk — Lifestyle changes and consider drug treatment	High Added Risk — Lifestyle changes and consider drug treatment	Very High Added Risk — Lifestyle changes and immediate drug treatment	Consider immediate hospitalization
Diabetes	Moderate Added Risk — Lifestyle changes for several months then drug treatment if BP uncontrolled	High Added Risk — Lifestyle changes and consider drug treatment	High Added Risk — Lifestyle changes and consider drug treatment	Very High Added Risk — Lifestyle changes and immediate drug treatment	Consider immediate hospitalization
Established Cardiovascular or Renal Disease	Very High Added Risk — Lifestyle changes and immediate drug treatment	Very High Added Risk — Lifestyle changes and immediate drug treatment	Very High Added Risk — Lifestyle changes and immediate drug treatment	Very High Added Risk — Lifestyle changes and immediate drug treatment	Consider immediate hospitalization

Adapted from Zedman S, et al. The Impact of an individualized risk-adjusted approach on hypertension treatment in primary care, 2017, Wiley; Norman M. Kaplan, New Guidelines for Hypertension Therapy from the USA, The Kelo Journal of Medicine 2009; NIH, Prevention, Detection, Evaluation, and Treatment of High Blood Pressure, 2002.

For the full attribution statement, please see the last page.

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Accessible version in your toolkit.

Blood Pressure (mmHg) Risk Assessment

Risk Factors or Disease	Normal: SBP under 120 and DBP under 80	Elevated: SBP 120-129 and DBP under 80	Stage 1 HTN: SBP 130-139 or DBP 80-89	Stage 2 HTN: SBP 140 and higher or DBP 90 and higher	Hypertensive Crisis: SBP over 180 or DBP over 120
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Established Cardiovascular or Renal Disease	Very High Added Risk — Lifestyle changes and immediate drug treatment	Very High Added Risk — Lifestyle changes and immediate drug treatment	Very High Added Risk — Lifestyle changes and immediate drug treatment	Very High Added Risk — Lifestyle changes and immediate drug treatment	Consider immediate hospitalization

Adapted from Zechmann S, et al. The impact of an individualized risk-adjusted approach on hypertension treatment in primary care, 2017, Wiley; Norman M. Kaplan, New Guidelines for Hypertension Therapy from the USA, *The Keio Journal of Medicine* 2009; NIH, Prevention, Detection, Evaluation, and Treatment of High Blood Pressure. 2003.

JNC 8 Indications for Blood Pressure Treatment

- Blood pressure goals and recommendations for therapy are based on age, race, diabetes, and chronic kidney disease.
- Recommendations:
 - **Diet and lifestyle therapies included in all stages of treatment.**
 - All ages with DM or CKD: goal BP <140/<90 mmHg.
 - >60 years without DM and CKD: goal BP <150/<90 mmHg.
 - <60 years without DM and CKD: goal BP <140/<90 mmHg.
- See the full JNC 8 practice guidelines in *JAMA* article:
“2014 Evidence-Based Guideline for the Management of High Blood Pressure in Adults: Report From the Panel Members Appointed to the Eighth Joint National Committee (JNC 8).”

Clinical Focus

- Summary, conclusion, and key takeaways.



ABCDs of the Functional Nutrition Evaluation



**Anthropometrics
& Vital Signs**



**Biomarkers &
Functional Labs**



**Clinical Findings
(Nutrition Exam)**



**Diet and Lifestyle
Assessment**

Three Ways to Define Metabolic Risk

- Body Mass Index (BMI)
- Weight/Height Ratio (WHtR)

Overweight or Obese



- Waist/Hip Ratio (WHR)
- Body Roundness Index (BRI)
- Waist Circumference (WC)

OverVAT

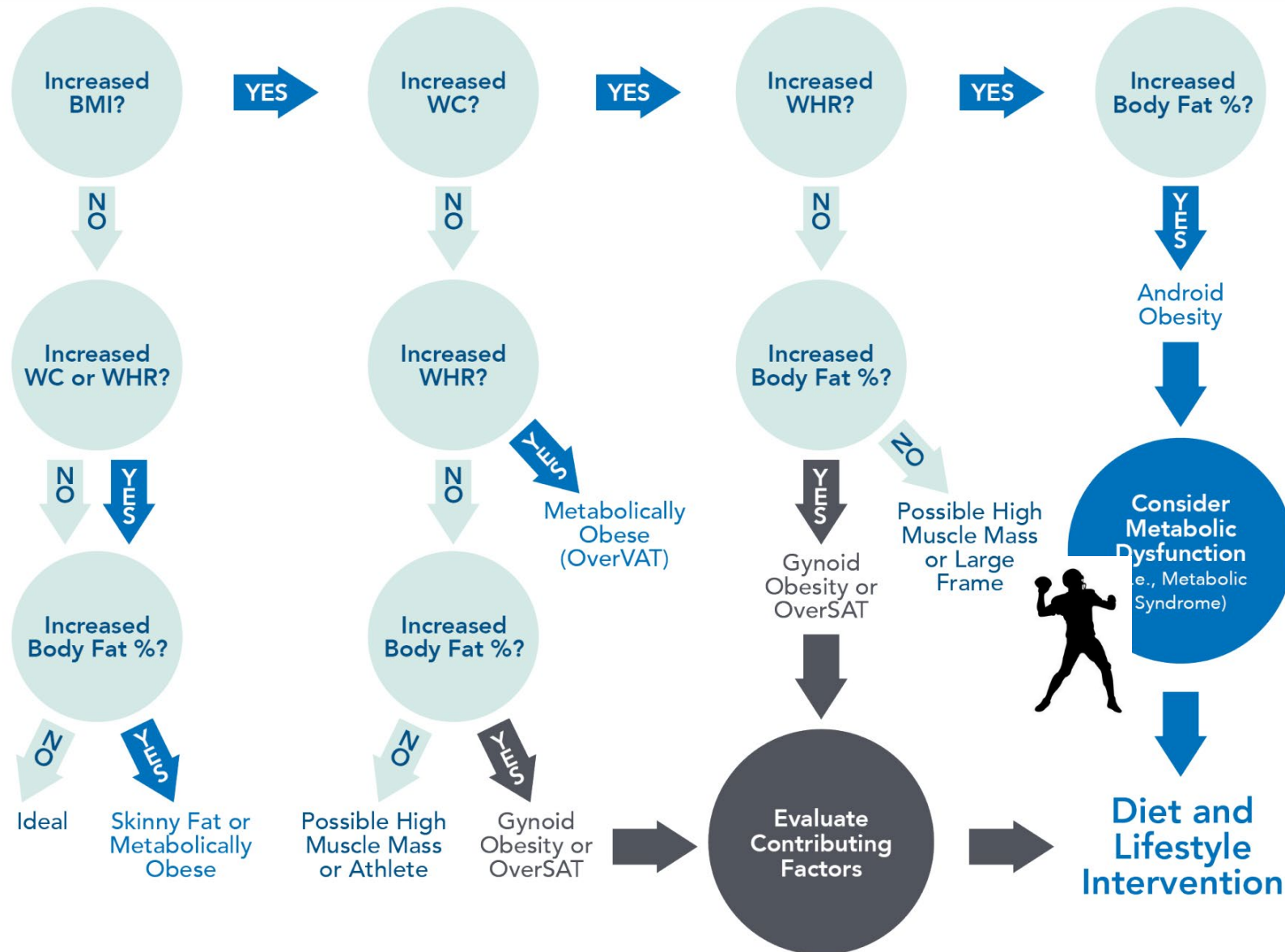


- Total body fat % may be estimated by other methods
- Bioimpedance Analysis (BIA)
- DEXA

OverFAT



Body Composition Assessment



Association of Nutrient Deficiencies With Hypertension

Proteins	Fats	Carbohydrates	Minerals	Vitamins	Phytonutrients
• Animal protein (lean or wild) • Plant protein	• Omega-3 fatty acids • Omega-9 fatty acids	• Soluble fiber	• Calcium • Magnesium • Potassium • Zinc	• Choline • Vitamin B6 • Vitamin C • Vitamin D	• Alpha lipoic acid • CoQ10 • Flavonoids • Lycopene • Low antioxidant profile

See: "References: Association of Nutrient Deficiencies With Hypertension"

References

ASSOCIATION OF NUTRIENT DEFICIENCIES WITH HYPERTENSION

- Houston MC. The role of nutrition, nutraceuticals, vitamins, antioxidants, and minerals in the prevention and treatment of hypertension. *Altern Ther Health Med*. 2013;19(Suppl 1):32-49.
- Houston M. Nutrition and nutraceutical supplements for the treatment of hypertension: part II. *J Clin Hypertens (Greenwich)*. 2013;15(11):845-851. doi:10.1111/jch.12212
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Key Clinical Takeaways



Anthropometric measurements are an efficient and effective means of determining cardiometabolic risk and the need for nutritional interventions.



Waist circumference and body roundness index are emerging as important measurements to determine visceral adiposity, which is arguably the most important risk parameter.

Key Clinical Takeaways



Body composition assessment can further inform level of risk beyond what is provided by other anthropometric measurements.



Elevated blood pressure is often a surrogate marker for nutrient deficiencies, and nutrient repletion may be an effective intervention to normalize blood pressure and reduce risk profile.

ABCDs of Nutrition Evaluation

Part A: Anthropometrics and Vital Signs

APPLYING FUNCTIONAL MEDICINE IN CLINICAL PRACTICE